

feature shares a linear relationship with any of the other feature inputs, its effects on the model output would yield no differences. Stick to variables that enhance the image of the dataset and provide better understanding of the model coefficients.

Decomposing some of the variables can yield features as well. If you were to take data values or any columns pertaining to time, the difference would give you a new feature – duration. You can also split such values into months, days, weekends and weekdays to give more depth to chronological data.

Categorical values, as discussed, would have to be split into separate columns for each of the values to generate good results. One should split the values into a sizeable range of ‘buckets’. In the air quality case, you can measure the concentration and insert a new column relating to ‘good’ or ‘bad’ air quality. These two values would become the buckets which will have their own columns. In data terminologies, this is known as hot encoding or dummy variable creation.

REACHING THE FINISH LINE

The final step to data preparation is to split the data into two sets - testing and training. Create non-overlapping subsets of your data for the training and evaluation sets. A rule of thumb states to split the data in a 60/40 or 70/30 fashion for the best results. If there’s ever any confusion over the best tools and softwares to be used for such operations, getting in touch with some coding examples from SQL and AWS are recommended.

The final step in the data preparation procedure looks at ensuring that all the values, whether categorical or numeric, are converted into a single format. Filtering the columns and rows for any redundancies and errors are given a final touch. If you’re handling such operations in Excel, use the Ctrl+F function or create a macro to scan the dataset for any red flags. The macro should be able to return the location of the cell and format it with reasonable changes.

For the large businesses and data centric strongholds, using a self-service data preparation software like Azure can be quite beneficial. What makes such cloud based softwares even more useful is their ability to leverage many attributes together without user interference. This also points towards another important need for data – pick a good workbench.

The larger the data set, the more difficult, as well as time consuming it is to clean. However, larger datasets generally provide better results downstream.